LASERCUBE

FIRMWARE v2.2
OCTOBER 2026

Firmware v2.2

106 new visualizers, every cube on your network from one page, keystone correction, and a live view of the laser.

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What's new

Everything new for LaserCube Ultra MK2 and Pro devices since firmware v2.0, including firmware v2.1, which came only on new cubes. The pages that follow show each new feature in detail.

106
New visualizers
Halloween figures, 3D dancers, new originals and classic LaserOS effects.

All
Your cubes, one page
Control every cube on your network from one browser tab.

Live
Laser preview
See the laser output on your phone as the audience sees it.

±45°
Keystone correction
A square picture on a wall at an angle.

Web control page

- Every cube on your network in one page, each on its own tab.
- Every picture setting in one popup: Image, Colour, Safety and Test.
- A live preview of the laser output.
- The audio input with a live spectrum, and a BPM tab with Tap BPM and Sync Beat.
- Changes show at once and are kept when you press Save.

Visualizers

- 21 Halloween figures and 10 dancing 3D characters, each dancing to the music.
- 19 new originals, from Tesseract to Galaxy Arms.
- 56 classic LaserOS effects, now built into the cube.

On the cube

- Keystone correction: square up the picture on a wall at an angle, up to 45° each way.
- Change the network mode, IP settings and WiFi details without a restart.
- A clock and a laser diode hours counter on the 3.0 PRO and 6.0 PRO.

Improvements

- Lighter rendering: visualizers use much less processor time.
- Full-colour content now reaches full laser power.
- A failed update keeps the previous firmware.
- Many stability fixes, from MIDI controllers to the SD card.

Your LaserCube, from your phone

The cube's web page now controls every cube on your network, and covers the whole picture setup and the playlist settings. Open it in your phone's browser: your cubes are the tabs along the top, and the buttons along the bottom open the picture and settings popups.



Your cubes
Each cube on your network gets a tab, named by its model. Tap one to control that cube.

Rescan
Looks for cubes on the network again.

Visualizers
Tap a preview to play it. The new visualizers are all here.

Status bar
WiFi signal and channel, temperature and battery.

Effects
Ready to use as soon as you connect, and closes with a tap outside it.

Settings
Playlist or Visualizer, Audio, BPM and General.

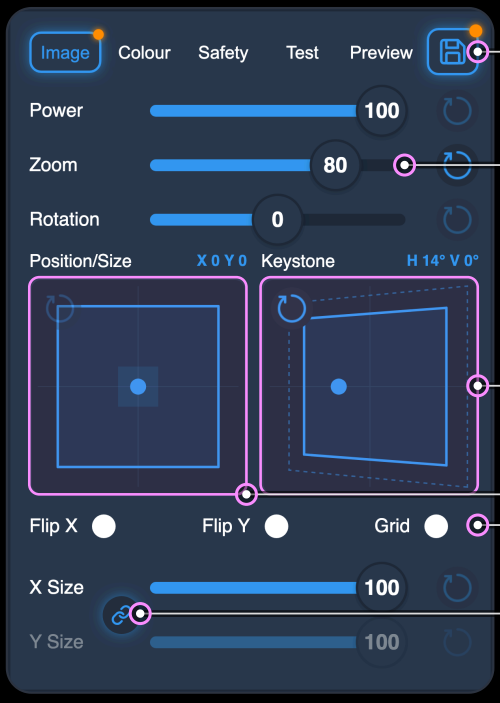
Picture settings
A new picture icon. Opens the Image, Colour, Safety, Test and Preview tabs.

Picture settings

All of the cube's projection settings in five tabs. Changes show on the laser straight away and the cube's own screen follows. Press Save to keep them: an orange dot marks a tab with unsaved changes.

Image

Power, Zoom and Rotation, the picture's position and size, keystone, flips and the grid. The two pads draw the picture as it lands on the wall.



Save
Keeps the changes on every tab. The dot means some aren't saved yet.

Power, Zoom and Rotation

Keystone pad
Drag to square up the picture on a wall at an angle.

Position/Size pad
Drag to move the picture. Pinch with two fingers to zoom.

Flip X, Flip Y and Grid
Grid projects a square to line things up.

Lock X-Y
Keeps Y Size equal to X Size.

Live preview

The laser output, live, as the audience sees it, with every picture setting and any image and colour effects applied. Set the size, position, keystone and safety zone while you watch the result.



One phone at a time
Another phone shows a message and takes the preview over with a tap. The cube's own Preview closes while a phone has it.

Live laser output
Drawn as the cube projects it, image and colour effects and flips included.

Colour

The cube's Colour Balance, from your phone.



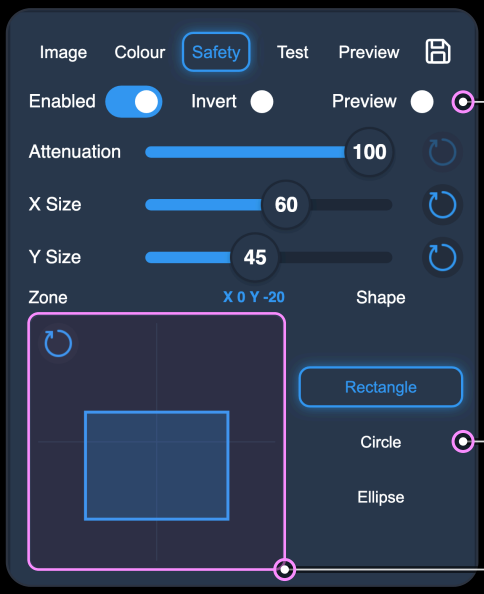
Max R, G and B
The most power each colour is driven at.

Min R, G and B
The least power a lit colour gets.

Gamma
The brightness curve.

Safety

The safety zone, the area the beam may draw in. Drag it on the pad or pinch it to resize, and turn on Preview to see its outline on the wall.



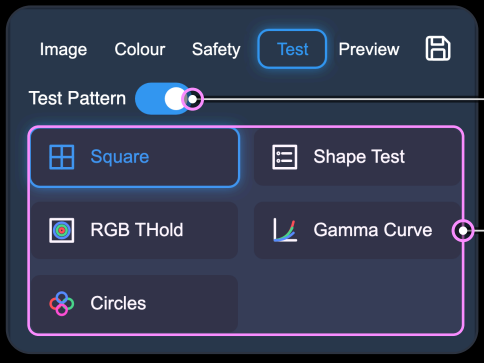
Enabled, Invert and Preview
Turn the zone on, keep the beam outside it instead, or show its outline on the wall until you close the popup.

Shape
Rectangle, circle or ellipse.

Zone pad
Drag to move the zone. Pinch to resize it.

Test

Project a test pattern straight from your phone. Every pattern the cube has gets its own button.



Test Pattern
Turns the pattern on and off.

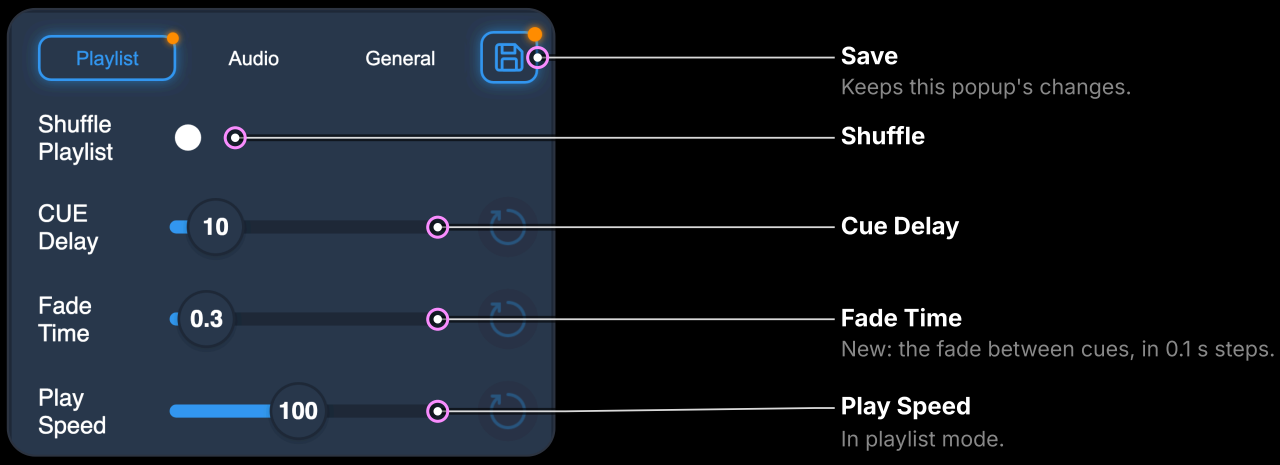
Patterns
Tap one to project it.

Settings

The settings popup has four tabs and its own Save button. In visualizer mode the first tab is called Visualizer, and the BPM tab appears.

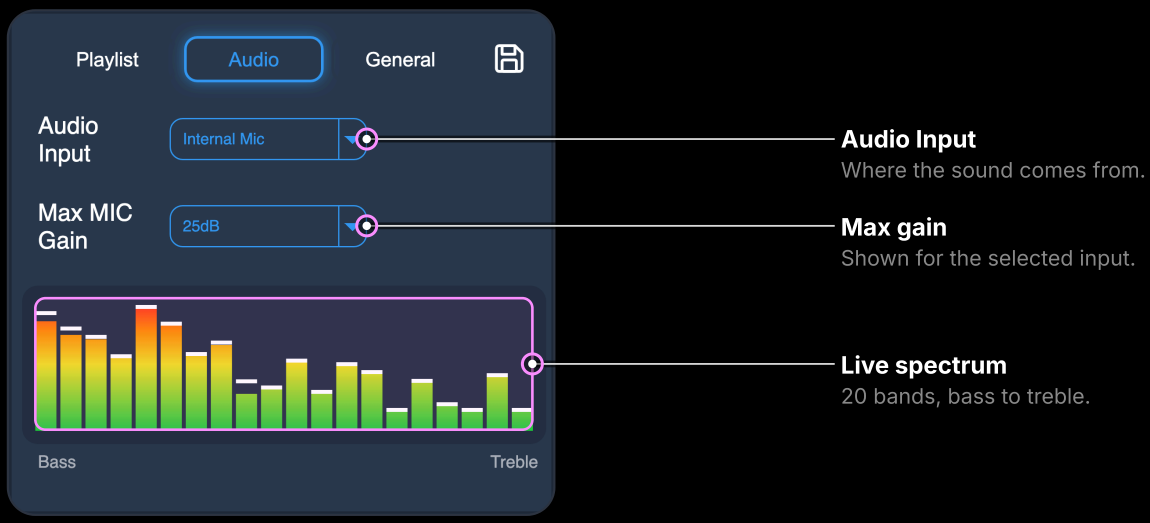
Playlist

Shuffle, Cue Delay, Fade Time and Play Speed. Fade Time is new on the web page.



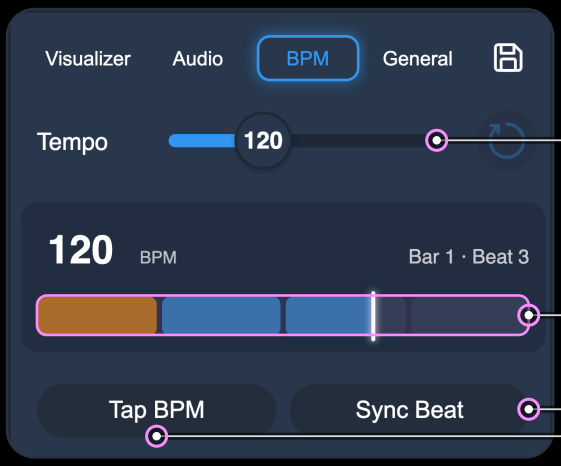
Audio

Choose the audio input and its gain, and see what the cube hears. Bars that sit in the red mean the gain is too high.



BPM

In visualizer mode: set the tempo the visualizers move to, watch the beat, and lock the cube to the music.



Tempo
60 to 250 BPM.

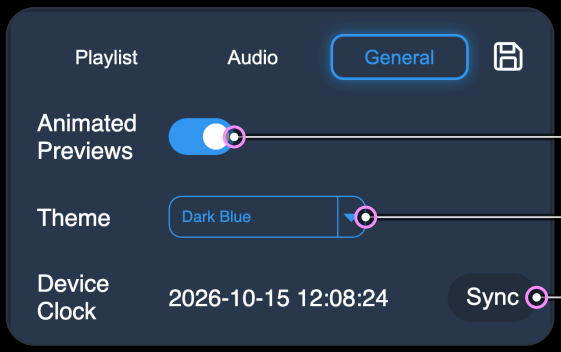
Beat display
The BPM pulses on every beat. One block per beat, beat 1 in orange, with a cursor sweeping the bar.

Sync Beat
Press on beat 1 to line the cube up with the music.

Tap BPM
Tap along to set the tempo. Your phone times the taps, so WiFi delay doesn't count.

General

Animated Previews, the page's colour theme, and the cube's clock.



Animated Previews
In playlist mode.

Theme
Six colour themes for the page.

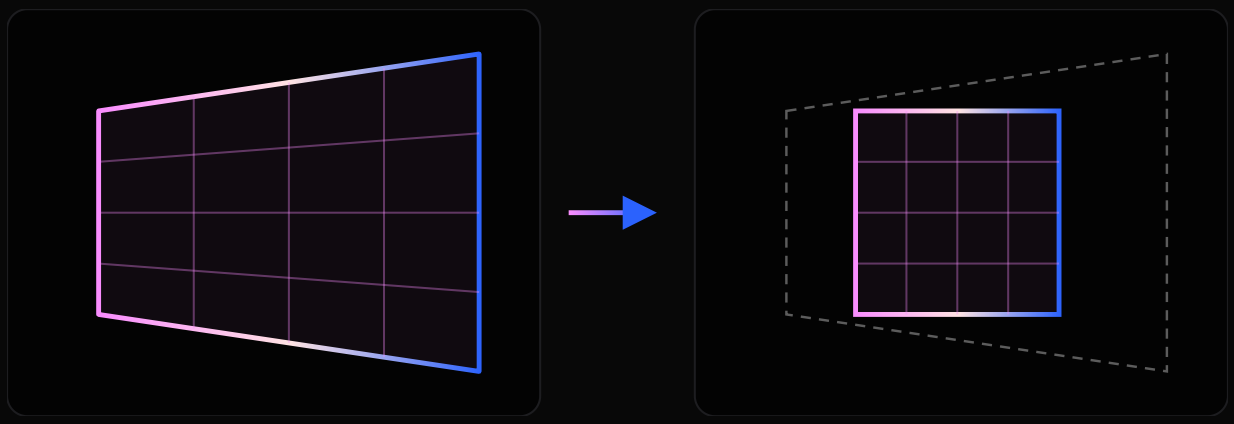
Device Clock
Sets the cube's clock from your phone, on cubes with a clock.

Made for phones and tablets

- Checked on 17 iPhones, Android phones and iPads, recent and older.
- Popups scroll on short screens and sit centred on tablets and computers.
- Tap outside a popup to close it.
- Reconnects faster after a cube restarts, and the previews load more smoothly.
- Buttons, switches and sliders follow your chosen theme colour.
- Reset buttons grey out when a setting is already at its default.
- Holding a button or slider no longer selects text or opens the long-press menu.

Keystone correction

Projecting onto a wall the cube isn't facing straight on turns a square into a trapezoid. Keystone correction reshapes the picture so it lands square, horizontally, vertically or both, up to 45° each way.



The cube at an angle to the wall: the square lands as a trapezoid.

Keystone set to the wall's angle: the square lands square.

Squaring up the picture

1. On the cube, open `Settings > Projection > Image Setup`.
2. Turn on Show Grid: the cube projects a square.
3. Turn Keystone H and Keystone V until the square looks square on the wall.
4. If the square comes out a little wide or narrow, adjust X Size or Y Size.

Good to know

- The picture gets a little smaller so it stays inside the scan area: about 90% of full size at 15°, 83% at 30° and 77% at 45°.
- Size, position and rotation still act on the picture as the audience sees it, and safety zones stay put in the room.
- The web page's Keystone pad does the same from your phone.
- Image Setup has an icon for every control, and Defaults now resets Zoom and keystone too.

Network, clock and more

Network changes without a restart

- Change the connection mode in `Settings > Connection`, or the IP settings and WiFi names and passwords under `Settings > Network`. The cube switches over in a few seconds, without restarting.
- The status bar shows the switch, then the new IP address.
- Changing the control mode still restarts the cube.

Clock on the 3.0 PRO and 6.0 PRO

- Set the date and time in `Settings > Date & Time`, or sync the clock from your phone on the web page.
- `Status Info` shows the date and time, and how many hours the laser diode has been used.
- Files the cube writes to the SD card now carry the real date.

Several cubes

- One cube's web page controls every cube on the network, from the tabs at the top.
- The cube serving the page needs to be in WiFi Server or LAN Server mode, with the other cubes joined to it as clients, in Playlist or Visualizer mode.
- A cube in WiFi Server mode now takes up to 8 phones, computers and cubes at once.
- A cube set to WiFi Client with no WiFi details of its own joins a cube in WiFi Server mode straight out of the box.

New model names

- The 3.0 EVO MK2 and 6.0 EVO MK2 are now the 3.0 PRO and 6.0 PRO, and the 1.2 PRO MK2 is the 1.2 PRO.
- Cubes take their new name when they are updated, on the screen and the web pages.

106 new visualizers

Halloween figures, 3D dancers, new originals, and the classic LaserOS effects, now built into the cube.

Halloween 21

Each one dances to the music with its own moves, and settles into a quiet pose when the music stops.



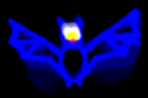
Spider 2D 2D

A purple spider on a silk thread, riding up and down it, swinging and spinning to the music.



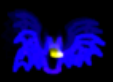
Spider 3D 3D

The same spider built from solid shapes, seen in three-quarter view.



Bat B1 3D

A bat that flaps, swoops and barrel-rolls, and sleeps upside down when the music stops.



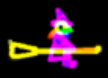
Bat B2 3D

The same bat drawn as one continuous silhouette line, with big round eyes.



Witch 2D 2D

A witch on a broomstick, hat and cape streaming, cackling on the strong beats.



Witch 3D 3D

The same witch in solid shapes, looping and diving through depth.



Black Cat 2D 2D

A soft cat that sways, hisses and prowls, and grooms its paw in the quiet.



Black Cat 3D 3D

The same cat in three-quarter view, with jointed legs and a long tail.



Vampire 2D 2D

Dracula sweeps his cape open on the bass and bursts into a swarm of bats on a drop.



Vampire 3D

3D

A vampire whose cape, purple outside and red inside, wraps forward like wings.



Mummy 2D

2D

A stiff, lurching mummy whose bandage unravels and whips round him.



Mummy 3D

3D

A shuffling mummy with bandages wound round solid limbs.



Thing 2D

2D

A disembodied hand that walks on its fingertips, taps, snaps and waves.



Thing 3D

3D

The same hand in solid shapes, doing handsprings, backflips and pirouettes.



Slimer

2D

A green ghost that flies about, chomps to the bass and eats from his hands.



Zombie Z1

3D

A shambling zombie dancing ten moves.



Ghost G1

3D

A sheet ghost that hovers, swoops and fades in and out, banking as it moves.



Pumpkin P1

2D

A candle-lit jack-o'-lantern on a curly vine that headbangs and cackles; its head can spin or pop off.



Pumpkin P2

2D

A pumpkin-headed stick dancer with 17 moves, its grin following a voice.



Skull S1

3D

A 3D skull with expressive brows, red eyes that keep eye contact, and a talking jaw.



Dancing Bones

3D

A full skeleton dancing 12 moves locked to the beat, standing and breathing in the quiet.

Dancing 3D characters 10

Full 3D dancers that keep time with the music and stand idle when it stops.



Dancing W1 3D

A dancer with 11 moves, including a leap, who breathes idly when the music goes quiet.



Dancing Man 3D

A streetwear dancer with disco points, hip swings, twists and shuffles.



Dancing W2 3D

A dancer with a bob haircut, whose moves include a moonwalk and a spin.



Dancing W3 3D

A dancer in a black dress, working through 15 moves on the beat.



Dancing MJ 3D

A dancer in a low fedora, short jacket, cropped trousers and loafers.



Dancing Bear 3D

A teddy bear with round paws, dancing on its short legs.



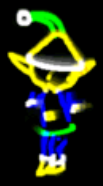
Dancing Alien 3D

A grey alien with a big egg-shaped head and long stick limbs.



Dancing Bird 3D

A crow-like bird dancing with its wings for arms.



Dancing Elf 3D

A Christmas elf in a floppy hat with a pom-pom, and pointed ears.



Dancing Santa 3D

Santa Claus with a round belly, flowing beard and fur-topped boots.

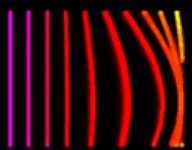
New originals 19

Abstract, beam and 3D visualizers made for the cube, each moving with its own part of the music.



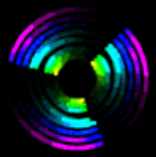
Lituus

A spiral swept from centre to rim in rotating kaleidoscope copies, turning with the beat.



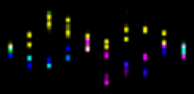
Op Art Warp

An op-art field of lines bent by a travelling bulge that deepens with the bass.



Orbital Arcs

A sphere of orbiting arcs pulsing out in shells, one for each part of the music.



Sine Bands

Mirrored rails of dotted sine waves, each band swelling with its own part of the music.



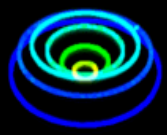
Speaker Dots

Rows of dots pushed outward by the sound like a speaker cone, the colours cycling faster when louder.



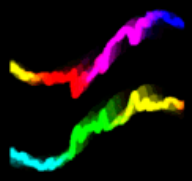
Plankton Storm

Drifting clouds of plankton, each lit by its own frequency, shimmering with short tails.



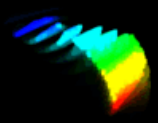
Ripple Dome

Rings ripple over a dome: the bass raises the waves and the treble speeds them up.



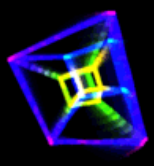
Sound Ribbon

A wide ribbon snaking across the frame, its width traced by the bass.



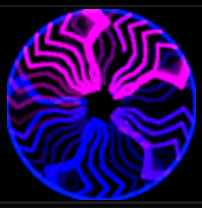
Spiral Stair

A spiral staircase round an empty core, each step driven by the music.



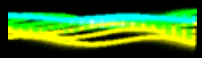
Tesseract

A 4D hypercube turning through its extra dimension, seen as two wireframe cubes trading places.



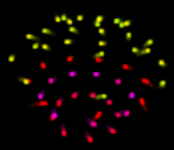
Cyclone Eye

Hurricane arms spiral round a calm eye that opens and closes with the volume.



Ocean Swell

Wave crests roll toward you, each wave's height set by part of the music.



Phyllo Sphere

A dotted sphere tumbling on every axis as ripples roll pole to pole and the bass swells it.



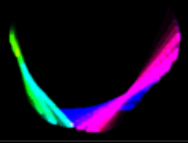
Starling Cloud

A starling murmuration wheeling through space, stretching on the bass.



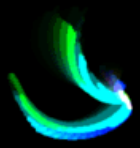
Cassini Split

Nested loops that pinch and tear into triangles of lobes as the music gets louder.



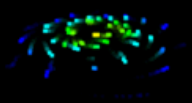
Chord Cardioid

String-art chords forming morphing cardioid shapes in two counter-rotating webs.



Clifford Bloom

Strange-attractor orbits trailing long filaments, reshaping with the music.



Galaxy Arms

Spiral galaxy arms sweep from an empty core, and peaks fling stars across the frame.

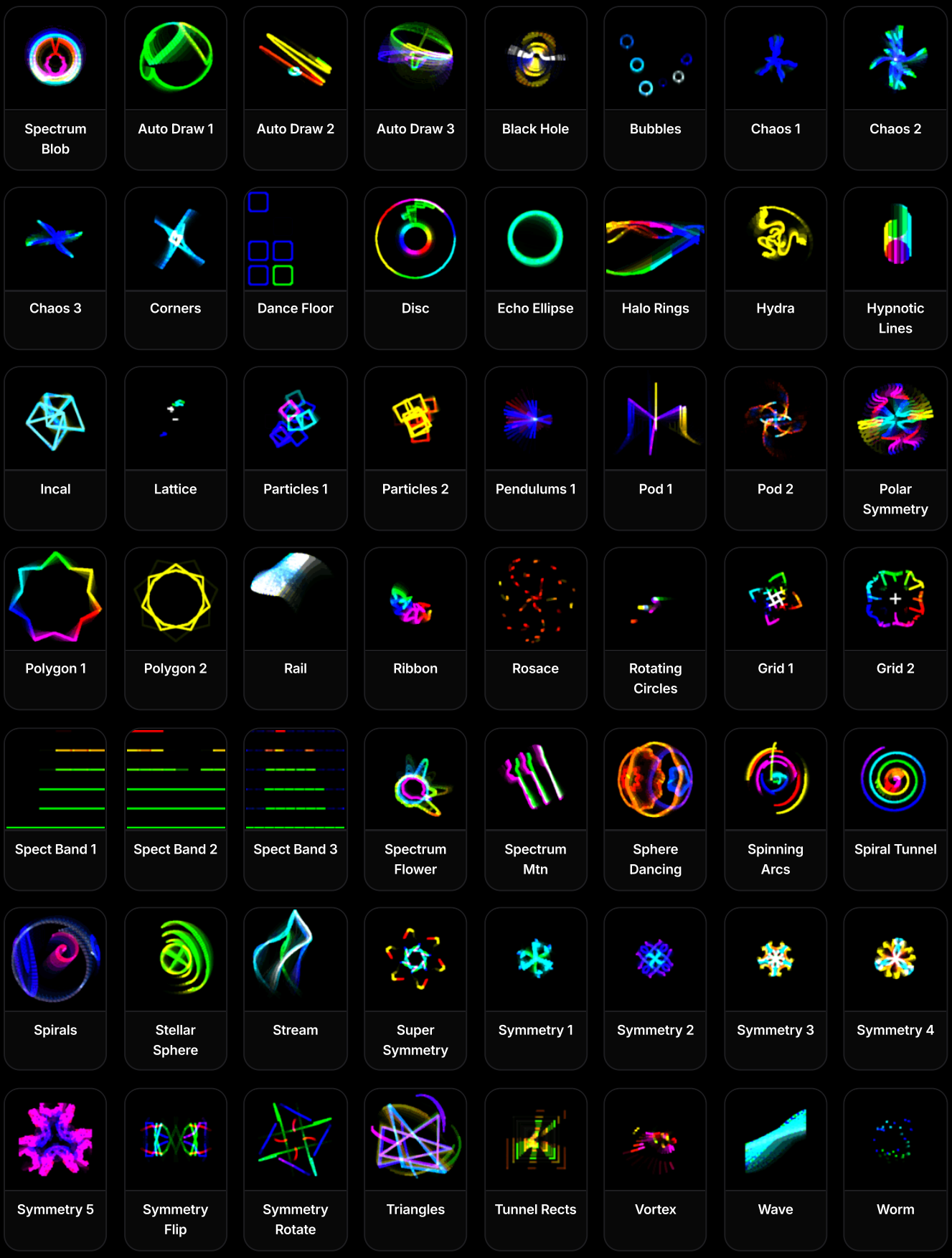


School Fish

A shoal of fish sweeping in banking formation, scattering on peaks and regrouping.

Classic LaserOS effects 56

The visualizers from the LaserOS app, now built into the cube, so they run on the cube itself.



Improvements and fixes

Performance and updates

- Lighter rendering: on the heaviest visualizers the render load dropped from about 69% to 52%.
- Full-colour visualizers and ILDA files now reach full laser power.
- If a web firmware update fails part way, the cube restarts into the firmware it already had and says so.

Network

- A cube running the network checks an address is free before handing it out, so fewer devices fail to connect.
- Changing the LAN Server IP no longer stops the cube handing out addresses until a restart.

On the cube

- Spinning the menu knob fast no longer skips steps or jumps backwards.
- The Colour Balance Min sliders show the true percentage.
- Starling Cloud no longer flashes a starburst as it starts.
- MIDI controllers can be unplugged and plugged back in: pads, lights and shortcuts reconnect.
- Removing the SD card while browsing visualizers is now handled smoothly.
- The screen recovers from display stalls, and popups no longer leave a grey bar.

Web pages

- Effects are ready as soon as you connect: the first connection after power-up turns them on with every intensity at zero, without changing your saved settings.
- Web pages are sent compressed, so they load faster.
- The browser no longer fills the WiFi boxes on the setup page with a saved login.

For streaming software developers

- LaserOS UDP protocol: `PING_WITH_ID` (0x28) returns the cube's ID, and `GET_FULL_INFO` reports whether the output is on and a packet-error count.
- `SET_ILDA_RATE` outside the supported range replies FAIL and keeps the current rate, and a failed 0x97 command saves nothing.
- A sample packet with a malformed length is dropped whole and counted, instead of playing.
- The protocol specification is updated to document version 1.1, with a complete example sender.

LASERCUBE

LaserCube is made by Wicked Lasers.
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